Introduction to Computer Science
Fall 2026
Homework 6

Due: November 8, 2026 at 23:58:59
No Copy-and-Paste. No Late Submission.

Instructions. Please write your answers to each problem using LaTeX and submit the compiled PDF file. If you are unfamiliar with LaTeX, you may refer to https://latex-tutorial.com/ or other tutorials for help.

If you find a problem difficult to understand or solve, you may use external resources, including search engines (such as Google), online references (such as Wikipedia), and AI tools (such as DeepSeek), to study relevant concepts and assist your learning. However, you are responsible for understanding your submitted solution, and **please make sure every word in your submission is written by yourself to avoid plagiarism**.

If you use any AI tool while working on the assignment, please include a short disclosure at the end of your submission stating which tool you used and briefly describing how you used it. Undisclosed use of AI tools may be subject to point deductions.

Gaussian toolbox. You may need these somewhere.

(1) If $x \sim \mathcal{N}(\mu, \sigma^2)$, we can sample x by sampling standard Gaussian noise $z \sim \mathcal{N}(0, 1)$, and setting $x = \mu + \sigma z$.

(2) If $x \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then $x + y \sim \mathcal{N}(\mu_1 + \mu_2,\ \sigma_1^2 + \sigma_2^2)$.

(3) $\mathrm{KL}\big(\mathcal{N}(\mu_1, \sigma^2 I) \,\|\, \mathcal{N}(\mu_2, \sigma^2 I)\big) = \frac{\|\mu_1 - \mu_2\|^2}{2\sigma^2}$.

1 Diffusion Transformer (5 pts)

Diffusion models are among the most powerful models for generating high-resolution images and videos. For image generation, we can iteratively add Gaussian noise to an image until the resulting distribution is approximately standard Gaussian. Conversely, we can view any image as the result of iteratively *denoising* a sample of pure Gaussian noise. This is what a diffusion model does: it learns how to denoise, and it learns this from the process of adding noise to images from the dataset.

The model forward process (adding noise). Let x_0 be a real image from the dataset. Then the forward process is a Markov process. For $t = 1, \ldots, T$,

$$q(x_t \mid x_{t-1}) = \mathcal{N}\big(\sqrt{1 - \beta_t}\, x_{t-1},\ \beta_t I\big),$$

where $\beta_1, \ldots, \beta_T$ is a fixed sequence of small numbers chosen by hand. Let $\bar{\alpha}_0 = 1$, $\bar{\alpha}_t := \prod_{s=1}^{t}(1 - \beta_s)$. The schedule is chosen so that $\bar{\alpha}_T$ is close to zero.

(a) Prove the closed form

$$q(x_t \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0,\ (1 - \bar{\alpha}_t) I\big).$$

The "true" denoising process. The parameters here are derived from the forward process with x_0 known, so this formula can be viewed as theoretical ground truth. Notice that here we use the same notation q, and don't confuse this with $q(x_{t-1} \mid x_t)$, which doesn't condition on x_0.

For $t = 2, \ldots, T$,

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t I), \qquad \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\, \beta_t,$$

$$\tilde{\mu}_t = \frac{\beta_t \sqrt{\bar{\alpha}_{t-1}}\, x_0 + (1 - \bar{\alpha}_{t-1})\sqrt{1 - \beta_t}\, x_t}{1 - \bar{\alpha}_t}.$$

The model denoising process. From (a) we know, for a fixed t, we can sample x_t directly as

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),$$

where ϵ is independent of x_0. A neural network $\epsilon_\theta(x_t, t)$ looks at the noisy image x_t and the time step t, and predicts this noise ϵ. One denoising step of the model is a Gaussian

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(\mu_\theta(x_t, t),\ \tilde{\beta}_t I\big), \qquad \mu_\theta(x_t, t) = \frac{1}{\sqrt{1 - \beta_t}}\Big(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\, \epsilon_\theta(x_t, t)\Big),$$

where $\tilde{\beta}_t$ is the same fixed variance defined above.

(b) Our training objective is to predict the noise contained in the image at each timestep. Intuitively there can be another training objective, that is, to fit the true denoising distribution with a Gaussian. Prove that this objective is proportional to our objective, i.e.

$$\mathrm{KL}\big(q(x_{t-1} \mid x_t, x_0) \,\big\|\, p_\theta(x_{t-1} \mid x_t)\big) = \frac{\beta_t^2}{2\tilde{\beta}_t(1 - \beta_t)(1 - \bar{\alpha}_t)}\, \|\epsilon - \epsilon_\theta(x_t, t)\|^2.$$

The other component of a diffusion transformer is the transformer, which we learned about in a previous homework. It is a really cool idea to combine diffusion, as an algorithm, with the transformer, as a model structure; and this hybrid is now widely used in video generation.

DiT often operates on autoencoder **latents** to reduce computation. A fixed autoencoder first compresses images into latent representations. Similar to tokenizing text, the latents are cut into small **patches**, and each patch is one token.

A practical transformer contains many engineering details. Here we wrap all these details in a single box, and treat the transformer as a machine that eats a sequence of tokens and produces a sequence of tokens of the same length. The output tokens are processed to be the predicted diffusion noise.

A Simple Video DiT. Consider a simplified video diffusion transformer which treats a video as one big 3-D block: F frames of size $H \times W$. The video autoencoder compresses time by $4\times$ and each spatial dimension by $8\times$; the DiT then cuts the compressed block into spacetime patches of size $2 \times 2 \times 2$ (each patch = 2 consecutive latent frames $\times$ 2 $\times$ 2 latent pixels). Each patch is one token, and every token can attend to every other token.

Then for a 4-second, 256×256, 24 fps video,

(c) How many tokens does the video become?

(d) In the full spacetime attention, each of the N tokens attends to all N tokens. A cheaper factorized design splits attention in two: each token first attends only to the tokens at the same time index, and then only to the tokens at the same spatial position across time. Compute the total attention cost of both designs C_{full} and $C_{\text{factorized}}$.

(e) Could we avoid all this by generating every frame independently, each with the same image model? Explain in one or two sentences what would go wrong.

2 World Action Models (2 pts)

A **world action model** is a learned simulator that couples *seeing* and *acting*. We feed it observations v_t and actions a_t; it predicts what happens next and tells us what to do. There are three natural paradigms for generating world action models.

- **Action-first:** decide or sample the action first, then generate the video of what would happen.
- **Video-first:** generate or imagine the video first, then infer which action explains it.
- **Joint:** predict the action and video together.

Below are four systems from the literature. For each of them, decide which paradigm it follows **at its core**, and justify your answer in one or two sentences.

(i) Learning Universal Policies via Text-Guided Video Generation [1].

(ii) Video Prediction Policy: A Generalist Robot Policy with Predictive Visual Representations [2] (This paper is from IIIS).

(iii) FACT: Failure-Aware Causal Training for World-Action Models [3].

(iv) World Action Models are Zero-shot Policies [4].

3 Tactile Sensing (3 pts)

Cameras are everywhere, and they all speak the same language: a grid of RGB numbers, gathered from internet-scale data. Tactile sensors are the opposite: they are hard to build, hard to calibrate, and no two designs even speak the same language.

The membrane model. A simple tactile skin can be made of an elastic membrane stretched with tension T over a length L, with both ends fixed. When an object presses on it, the membrane deflects by $u(x)$. For small deflections and some other fine assumptions, force balance on a tiny piece of the membrane yields

$$T\,u''(x) = -q(x), \qquad u(0) = u(L) = 0,$$

where $q(x) \geq 0$ is the contact force per unit length, which is the thing the sensor ultimately wants to know, and $u(x)$ is what it actually measures (for example, a tiny camera watches the membrane, as in vision-based tactile sensors).

(a) Suppose the load contains a component at wavelength λ: $q_\lambda(x) = A\sin(2\pi x/\lambda)$. Solve the ODE to find the corresponding component of the deflection.

(*Hint:* Don't get scared. You must know the form of sine derivatives. Guess the formula, add the general solution of $u'' = 0$, and fix it with the boundary conditions.)

By the way, by *component*, we mean that any square-integrable function on $[0, L]$ can be viewed as a series of components of sine and cosine functions.

(b) The camera measurement of u can be noisy. Let the amplitude of camera noise $\delta = 1$ μm, wavelength $\lambda = 0.2$ mm and tension $T = 2$ N, calculate the amplitude of the error of the recovered force.

(c) With the same δ, λ and T as in (b), let $L = n\lambda/2$ for a positive integer n. Consider two nonnegative load densities satisfying

$$q_b(x) = q_a(x) + 1 \text{ N/mm} \cdot \sin(2\pi x/\lambda)$$

Calculate the maximal difference between two deflections. Can the camera tell them apart effectively?

Since such a tactile detector faces a lot of problems, people have designed many different tactile sensors. However, each of them has different headaches. Here is a list of some common sensor types. A taxel means a tactile pixel.

#	Sensor type	Typical information	Main limitations
1	Piezoresistive array	Normal pressure per taxel	Hysteresis, drift, array crosstalk
2	Capacitive array	Normal pressure per taxel	Parasitic capacitance, crosstalk, readout complexity
3	Piezoelectric (e.g. PVDF)	Dynamic force or vibration	Poor static sensing due to charge leakage
4	Magnetic (magnet + Hall sensors)	Normal and shear force estimates	Magnetic interference, calibration
5	Vision-based (e.g. GelSight, DIGIT)	Contact geometry; marker motion if present	Elastomer wear; force inference requires calibration or modeling
6	Electrical impedance tomography (EIT)	Contact map inferred from boundary electrical measurements	Ill-posed reconstruction, limited spatial resolution

(d) For each scenario below, choose the better-suited option from the two sensor types given, using the capabilities and limitations in the table. Justify your choice in one or two sentences. Assume that each sensor is properly calibrated.

(i) **Sustained grasp force control.** A gripper must monitor a steady normal contact force during a long, motionless hold. Choose between a **capacitive array** and a **piezoelectric sensor**.

(ii) **Slip-risk monitoring.** A controller needs both normal and shear contact-force estimates to assess slip risk. Choose between a **magnetic sensor** and a **piezoresistive array**.

(iii) **Fine texture recognition.** A robot must resolve fine surface ridges while maintaining stationary contact. Choose between a **vision-based tactile sensor** and an **EIT sensor**.

Declaration of AI Use

Please disclose any use of generative AI tools in completing this homework. For each tool used, briefly state what you used it for. You remain responsible for the correctness and originality of your submission. If you did not use any generative AI tools, remove this section.

References

[1] Yilun Du, Sherry Yang, Bo Dai, Hanjun Dai, Ofir Nachum, Josh Tenenbaum, Dale Schuurmans, and Pieter Abbeel. Learning universal policies via text-guided video generation. *Advances in neural information processing systems*, 36:9156–9172, 2023.

[2] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. In *Forty-second International Conference on Machine Learning*, 2025.

[3] Quanquan Peng, Yutong Liang, Rui Yan, Nicklas Hansen, and Xiaolong Wang. Fact: Failure-aware causal training for world-action models. In *Conference on Robot Learning (CoRL)*, 2026.

[4] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, Ayaan Malik, Kyungmin Lee, William Liang, Nadun Ranawaka, Jiasheng Gu, Yinzhen Xu, Guanzhi Wang, Fengyuan Hu, Avnish Narayan, Johan Bjorck,

Jing Wang, Gwanghyun Kim, Dantong Niu, Ruijie Zheng, Yuqi Xie, Jimmy Wu, Qi Wang, Ryan Julian, Danfei Xu, Yilun Du, Yevgen Chebotar, Scott Reed, Jan Kautz, Yuke Zhu, Linxi "Jim" Fan, and Joel Jang. World action models are zero-shot policies, 2026.